

SUNNY SHRESTHA

(720) 262-6825 - sunny.shrestha09@gmail.com - [linkedin.com/in/sunnyshrestha](https://www.linkedin.com/in/sunnyshrestha) - github.com/SunnyShrestha-repo - [Google Scholar: Publications](https://scholar.google.com/citations?user=...)

EXPERIENCE

Graduate Researcher | University of Denver | September 2021 – Present, US, CO, Denver

- Conduct multiple mixed-methodology studies focused on topics like Artificial Intelligence (AI), machine learning and human computer interaction (HCI).
- Employ NLP and statistical data analysis techniques using Python, R, and relevant libraries (Pandas, NumPy, SciPy, and Matplotlib) to generate insights from the studies.
- Publish peer-reviewed academic papers at renowned venues such as ACM SIGCHI, CSCW, and Frontiers in Artificial Intelligence, on the intersection of machine learning, AI, and user privacy.
- Develop algorithms iteratively for solving complex problems and improve them based on defined success criteria.
- Visualize data analysis findings and share with both technical and non-technical audiences.

Data & Innovation Analyst | City of Longmont | October 2023 – December 2025, US, CO, Longmont

- Automated manual data workflows, increasing efficiency and improving data reliability across City departments.
- Led migration of traffic crash data from flat files to ArcGIS geospatial datasets, directly supporting the City's Vision Zero initiative.
- Converted legacy and web-based data sources into structured, analysis-ready datasets.
- Built and operationalized Snowflake data warehouse instances to centralize data and enable scalable analytics.
- Delivered policy-driven dashboards and visualizations (Power BI) to support executive and policymaker decision-making.

Software Engineer & DevOps Engineer | Cloudmed Revenue Intelligence Solutions | January 2019 – July 2020, US

- Implemented data ETL process automation by administering weekly code deployment using Git and Docker, leading to a 72% reduction in human resource requirements.
- Managed Azure cloud services architecture to ensure efficient and reliable product deployment.
- Led the development and implementation of CI/CD pipelines and automated QA/QC practices.
- Streamlined, monitored, and enhanced systems such as AWS, PostgresDB, and fireworks across various platforms by using technical skills in infrastructure management and CI/CD tools.

Associate Software Developer | Acustream Inc | September 2016 – December 2018, US, CO, Lafayette

- Developed and implemented Python algorithms for data processing and invoicing, leading to an 84% increase in revenue generation.
- Created KPIs to assess the health of architecture components and the overall system, contributing to 66% increase in performance.
- Contributed to back-end and front-end development work in high-demand situations.
- Utilized project management tools, such as JIRA and Confluence, to efficiently track and update issues, and create standard operating procedures (SOP) for consistent and effective operations.

SKILLS

Programming Languages: Python, Java, C++, R (with a focus on NLP techniques)

Data Analysis and Visualization: Pandas, Matplotlib, Seaborn, MATLAB, Power BI, Tableau

ML Modeling: Logistic Regression, K-Nearest Neighbors, Random Forest, SVM, Cross-Validation, Feature Generation

Database: Postgres SQL, MS SQL, MySQL, MongoDB

ML Library: TensorFlow, Scikit-learn

Applications: Figma, Atlassian (JIRA, Confluence), Git, Gazebo, AWS, Slack

EDUCATION

MSc, Computer Science | University of Denver, Denver, CO | March 2023

Thesis: **Gender-Based Bias Mitigation Techniques in Algorithms for Music Recommender Systems**

- Implemented Collaborative Filtering mRS, bias mitigation techniques, and user study involving NLP methodologies in a sample of 20 college students.

BSc. Computer Information Technology | Minnesota State University- Mankato, MN | June 2016

RELEVANT PUBLICATIONS

- Exploring gender biases in ML and AI academic research through systematic literature review, Frontiers in AI 2022 (first author)
- SecureLD: Secure and Accessible Learning for Students with Disabilities, HFES 2022 (first author)
- SoK: An Evaluation of the Secure End User Experience on the Dark Net through Systematic Literature Review 2022
- Parents, Educators, & Caregivers Cybersecurity & Privacy Concerns for Remote Learning During COVID-19, ACM SIGCHI2021
- SoK: A Systematic Literature Review of Bluetooth Security Threats and Mitigation Measures, EISA 2021 (first author)